

Unit-4 - Virtualization and Capacity Planning

- ❖ Virtualization Technologies,
- ❖ Abstraction versus Virtualization
- ❖ Load Balancing and Virtualization,
- ❖ The Google Cloud
- ❖ Hypervisors,
- ❖ Virtual Machine Imaging,
- ❖ Porting Applications and
- ❖ Capacity Planning

Unit-4 - Virtualization and Capacity Planning

- ❖ Virtualization is a core technology in cloud computing that enables multiple virtual instances to **run on a single physical machine**, optimizing **resource utilization**, **scalability**, and **efficiency**.
- ❖ The different types of virtualization that are characteristic of cloud computing:
 - **Access:** A client can request access to a cloud service **from any location**.
 - **Application:** A cloud has multiple application instances and **directs requests to an instance** based on conditions.
 - **CPU:** Computers can be partitioned into a set of virtual machines with each machine being assigned a workload. Alternatively, systems can be virtualized through **load-balancing technologies**.
 - **Storage:** Data is stored **across storage devices** and often **replicated for redundancy**.

Load Balancing & Virtualization

- ❖ Cloud computing relies on virtualized network access, directing users to available resources.
- ❖ Load balancing is the technology that distributes service requests efficiently, improving performance.
- ❖ It can be implemented in hardware (e.g., F5's BigIP servers) or software (e.g., Apache mod_proxy_balancer, Pound, Squid).
- ❖ Load balancing optimizes resource use by increasing throughput, reducing latency, improving response times, and preventing system overload.

Load Balancing & Virtualization

- ❖ A load-balancing system can use different mechanisms to assign service direction.
- ❖ In the simplest load-balancing mechanisms, the load balancer listens to a network port for service requests.
- ❖ When a request from a client or service requester arrives, the load balancer uses a scheduling algorithm to assign where the request is sent.
- ❖ Typical scheduling algorithms in use today are round robin and weighted round robin, fastest response time, least connections and weighted least connections, and custom assignments based on other factors.

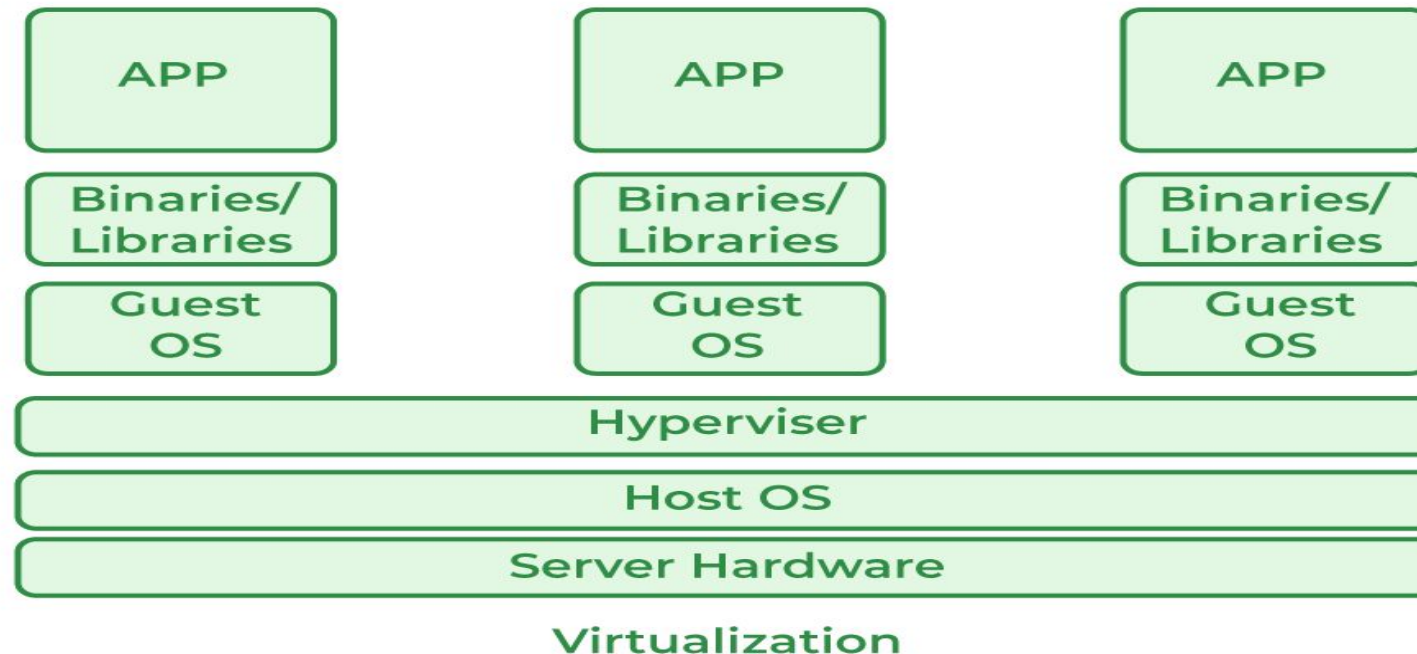
Advanced Load Balancing

- Advanced load balancers manage workloads by analyzing resource usage, response time, and queue length to distribute tasks efficiently.
- They offer features like health monitoring, traffic prioritization, security, and data compression.
- An Application Delivery Controller (ADC) enhances load balancing by integrating application server functions.
- It assigns a Virtual IP (VIP) to a server pool based on set rules and operates at both the network and application layers. Also called content or multilayer switches, ADCs provide services like content caching, SSL offloading, security, and advanced routing.
- They improve performance through network and application optimization, making them key for accelerating web services.

Virtualization in Cloud Computing

Definition:

- Virtualization in cloud computing is the process of creating virtual versions of computing resources, such as servers, storage, networks, and applications.
- It allows multiple virtual environments to run on a single physical system, improving resource utilization and efficiency.



Virtualization in Cloud Computing

- The term virtualization is often synonymous with **hardware virtualization**, which plays a fundamental role in efficiently delivering **Infrastructure-as-a-Service (IaaS)** solutions for cloud computing.
- Moreover, virtualization technologies provide a virtual environment for not only executing applications but also for **storage**, **memory**, and **networking**
- **Host Machine:** The machine on which the virtual machine is going to be built is known as Host Machine.
- **Guest Machine:** The virtual machine is referred to as a Guest Machine.

Virtualization in Cloud Computing

- Benefits of Virtualization:
 - ✓ More flexible and efficient allocation of resources.
 - ✓ Enhance development productivity.
 - ✓ It lowers the cost of IT infrastructure.
 - ✓ Remote access and rapid scalability.
 - ✓ High availability and disaster recovery.
 - ✓ Pay peruse of the IT infrastructure on demand.
 - ✓ Enables running multiple operating systems.

Virtualization in Cloud Computing

Drawback of Virtualization:

- ✓ **High Initial Investment:** Clouds have a very high initial investment, but it is also true that it will help in reducing the cost of companies.
- ✓ **Learning New Infrastructure:** As the companies shifted from Servers to Cloud, it requires highly skilled staff who have skills to work with the cloud easily and for this, you have to hire new staff or provide training to current staff.
- ✓ **Risk of Data:** Hosting data on third-party resources can lead to putting the data at risk, it has the chance of getting attacked by any hacker or cracker very easily.

Virtualization in Cloud Computing

Types of Virtualization in Cloud Computing:

- ✓ **Server Virtualization** – Divides a physical server into multiple VMs, each running its own OS and applications.
- ✓ **Storage Virtualization** – Combines multiple storage devices into a single virtual storage unit for better management and accessibility.
- ✓ **Network Virtualization** – Creates multiple virtual networks on a single physical network to enhance security and flexibility.
- ✓ **Desktop Virtualization** – Hosts user desktops on remote servers, allowing access from anywhere.
- ✓ **Application Virtualization** – Runs applications in a virtual environment, independent of the underlying OS.

Virtualization in Cloud Computing

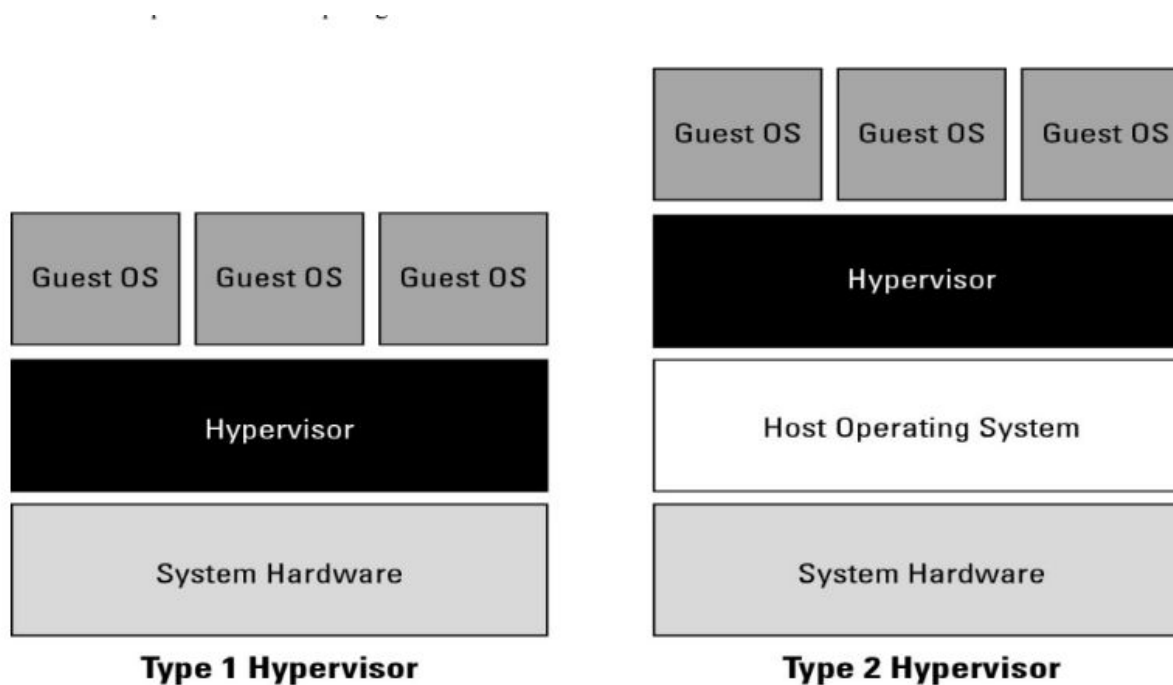
Hypervisors in Cloud Computing:

What is a Hypervisor?

- A hypervisor is a software that allows us to construct and manage virtual machines.
- Hypervisors are often referred to as Virtual Machine Monitors (VMMs).
- A hypervisor allows a single host computer to handle numerous guest virtual machines (VMs) by dividing resources such as memory and computation.
- When utilized as a hypervisor, the hardware is referred to as the Host, and the multiple VMs that utilize its resources are referred to as Guests.

Virtualization in Cloud Computing

Hypervisors in Cloud Computing:



Virtualization in Cloud Computing

Hypervisors in Cloud Computing:

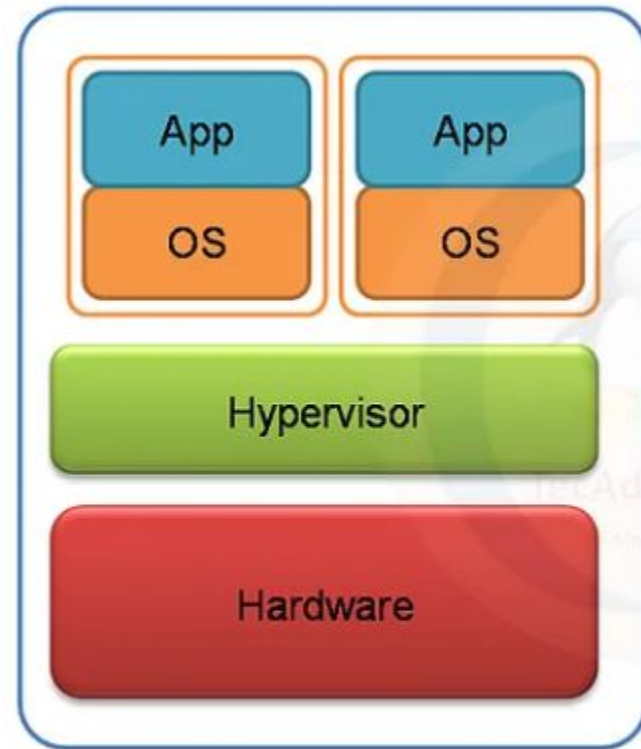
The hypervisor is categorized into two types:

1. **Type-1 Hypervisor:** - This runs directly on the host machine's physical hardware, so it's also known as a Bare-Metal Hypervisor.
2. **Type-2 Hypervisor:** – runs on the operating system of the physical host machine; hence, they are also called Hosted Hypervisors.

Virtualization in Cloud Computing

Type-1 Hypervisor:

- This hypervisor is installed directly on top of the physical server's hardware, there is no operating system or any other software layer in between. What we have in this case is:
 - ✓ A physical server/machine
 - ✓ Hypervisor installed on the hardware
 - ✓ Guest virtual machines



Type 1 (Bare Metal)
Virtualization

Virtualization in Cloud Computing

Type-1 Hypervisor:

- Type-1 hypervisors are actually a very basic OS on top of which we run virtual machines.
- So, the physical machine on which the hypervisor is running can only be used for virtualization purposes and nothing else.
- These hypervisors require a separate management console to perform activities like instance creation, migration, etc.
- The management console can be web-based or a separate package that you can install on another machine.
- Using this console, you can connect to the hypervisor to manage your virtual environment operations.

Virtualization in Cloud Computing

Type-1 Hypervisor:

Advantages:

1. **Better performance** – No extra OS layer consuming resources.
2. **Improved security** – Reduced attack surface compared to OS-based systems.
3. **Higher efficiency** – Direct access to hardware for optimized resource management.
4. This is why Type-1 hypervisors are commonly used in enterprise environments, where performance, scalability, and security are critical.
5. **Examples of Type-1 Hypervisors** – VMware vSphere with ESX/ESXi, Microsoft Hyper-V, Citrix Hypervisor (Xen Server).

Virtualization in Cloud Computing

Type-2 Hypervisor:

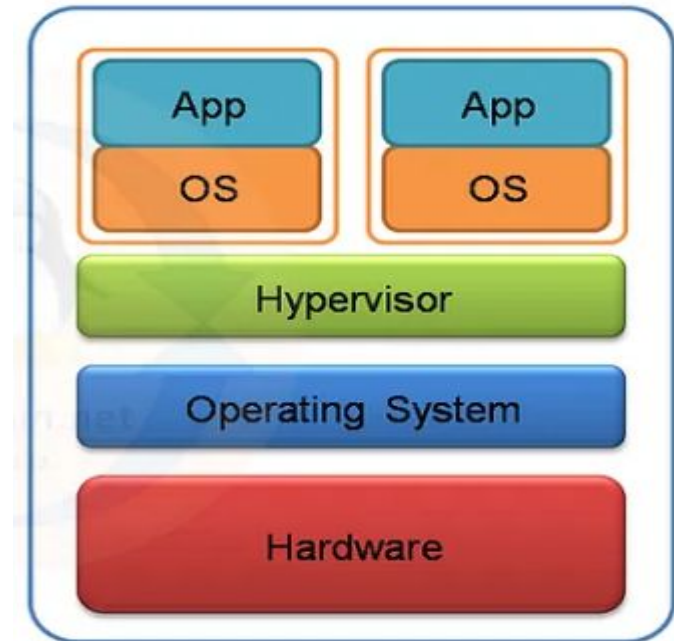
What is a Type-2 Hypervisor?

- A Type-2 hypervisor, also known as a **Hosted Hypervisor**, is software that **runs on top of a host operating system** (OS) to create and manage virtual machines (VMs).
- Unlike Type-1 hypervisors that run directly on hardware, Type-2 hypervisors **rely on the host OS** for resource management and device interaction. What we have in this case is:
 1. A physical server
 2. OS installed on that server hardware (OSes like Windows, Linux, macOS)
 3. Type-2 hypervisor on that OS
 4. Virtual machine instances/guest VMs

Virtualization in Cloud Computing

Type-2 Hypervisor:

- These hypervisors are usually used in environments where there are a **small number of servers**.
- They **do not need a separate management console** to create and manage the virtual machines.
- These operations can typically be **done on the server** that has the hypervisor hosted.
- This hypervisor is **treated as an application** on your host system.



Type 2 (Hosted)
Virtualization

Virtualization in Cloud Computing

Type-2 Hypervisor:

How Type-2 Hypervisors Work?

1. **Host OS Layer** – The hypervisor is installed as an application on a regular operating system (Windows, Linux, or macOS).
2. **Hypervisor Layer** – The hypervisor creates and manages virtual machines, assigning system resources like CPU, RAM, and storage.
3. **Guest OS Layer** – Each VM runs its own OS and applications, independent of the host OS.

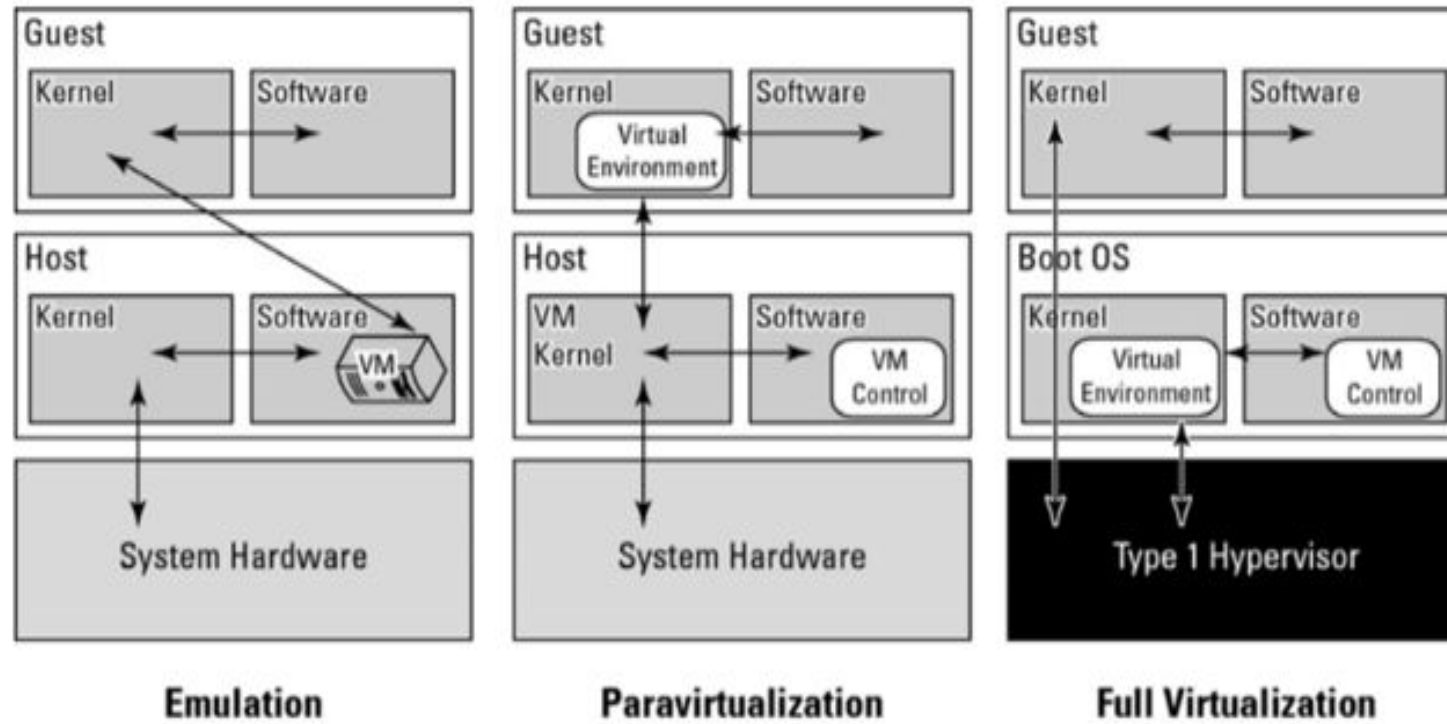
Virtualization in Cloud Computing

Type-2 Hypervisor:

Advantages of Type-2 Hypervisor:

1. **Simple management:** They essentially act as management consoles. There is no need to install a separate software package to manage the virtual machines running on type-2 hypervisors.
2. **Useful for testing purposes:** They are convenient for testing any new software or research projects. You can simply run multiple instances with different OS to test how the software works in each environment.
3. **Examples of Type 2 Hypervisors** – VMware Workstation Pro/VMware Fusion, Oracle VirtualBox, Microsoft Virtual PC, and etc...

Emulation Vs Full Virtualization Vs Para Virtualization



Emulation Vs Full Virtualization Vs Para Virtualization

Emulation:

- Emulation allows a system to **mimic another system's hardware or software** environment.
- It enables running applications designed for one platform on a completely different platform.
- Uses software to replicate hardware behavior.
- Slower due to instruction translation and processing overhead.

Example:

1. Running **Windows applications on macOS** using **Wine**.
2. Running **Android apps on a PC** using an **Android Emulator**.
3. Cloud **providers offering emulated environments** for legacy applications.

Emulation Vs Full Virtualization Vs Para Virtualization

Para Virtualization:

- In para virtualization, the guest operating system is aware of its virtualized environment and interacts with the hypervisor through special paravirtualized drivers.
- Paravirtualization eliminates the need for hardware emulation, resulting in lower overhead and improved performance compared to full virtualization.
- However, para virtualization requires modifications to the guest operating system to support paravirtualized drivers.
- **Examples:** Xen hypervisor with Xen-aware guest operating systems.

Emulation Vs Full Virtualization Vs Para Virtualization

Full Virtualization:

- In full virtualization, the guest operating system is unaware that it's running in a virtualized environment.
- The hypervisor presents a virtualized hardware environment to the guest operating system, mimicking the underlying physical hardware.
- Full virtualization provides broad compatibility with unmodified guest operating systems but may incur performance overhead due to hardware emulation.
- **Examples:** VMware ESXi with Intel VT-x/AMD-V hardware-assisted virtualization.

Emulation Vs Full Virtualization Vs Para Virtualization

FEATURE	EMULATION	FULL VIRTUALIZATION	PARA VIRTUALIZATION
Guest OS Modification	No	No	Yes
Performance	Slow	Moderate	High
Hardware Dependency	No	Yes	Yes
Hypervisor Used	Not needed	Type-1 or Type-2	Type-1
Use Case	Running software on different architectures	Cloud VMs, hosting multiple OSES	High-performance cloud VMs
Example	Running PlayStation games on a PC	VMware, VirtualBox, KVM	Xen, VMware ESXi

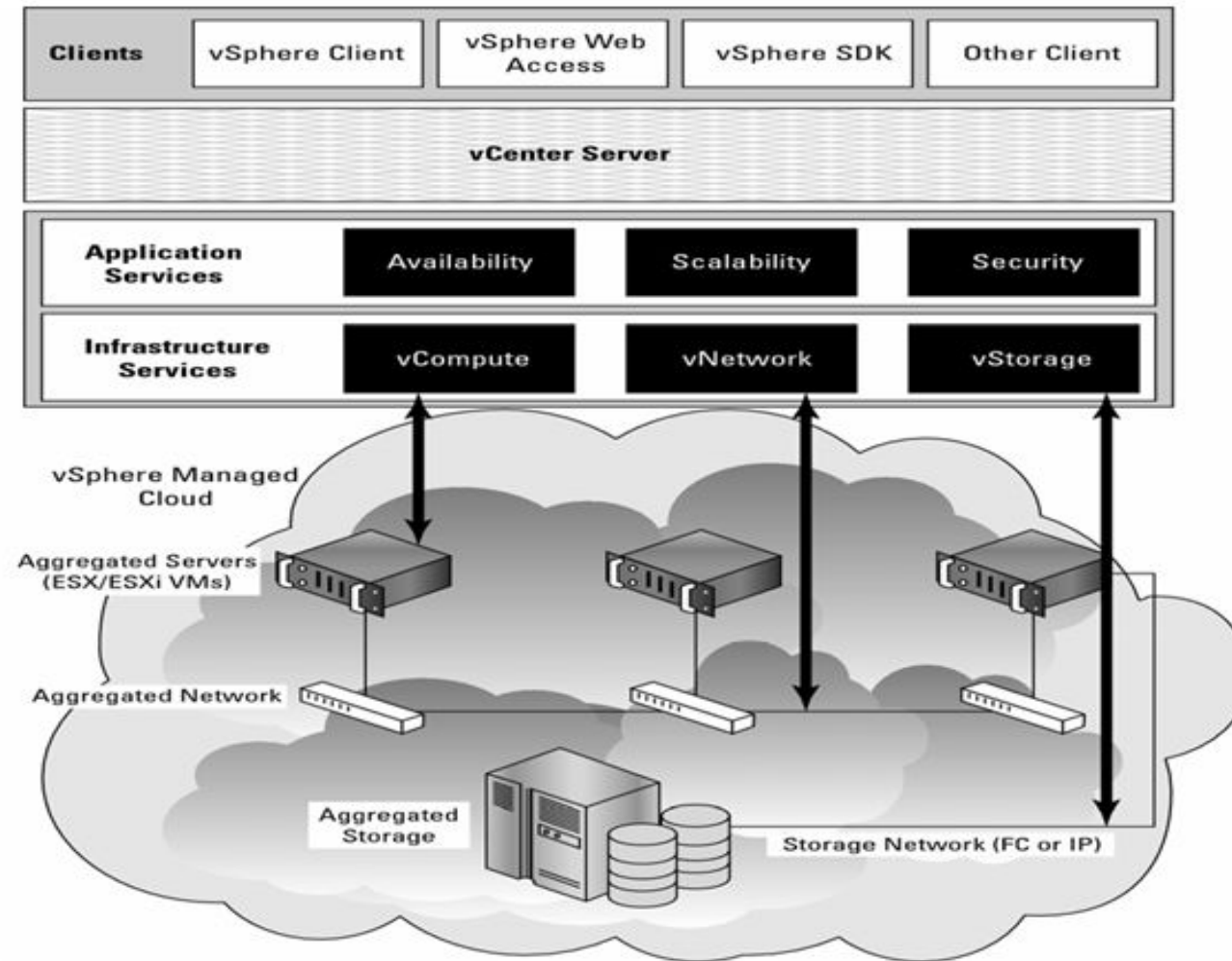
VMware vSphere

VMware vSphere

- **VMware vSphere** is a **Management Infrastructure Framework** that virtualizes **system, storage, and networking hardware** to create cloud computing infrastructures.
- **vSphere** is the **branding** for a **set of management tools** and a **set of products** previously labeled VMware Infrastructure.
- **vSphere** provides a set of services that applications can use to access cloud resources, including these:
 1. **VMware vCompute**: A service that aggregates servers into an assignable pool
 2. **VMware vStorage**: A service that aggregates storage resources into an assignable pool
 3. **VMware vNetwork**: A service that creates and manages virtual network interfaces
 4. **Application services**: Such as HA (High Availability) and Fault Tolerance
 5. **vCenter Server**: A provisioning, management, and monitoring console for VMware cloud infrastructures

VMware vSphere

VMware's vSphere cloud computing infrastructure model



VMware vSphere

- VMware is a very highly developed infrastructure and the current leader in this industry.
- A number of important add-on products are available for cloud computing applications.
- These are among the more notable products:
 1. **Virtual Machine File System (VMFS):** A high-performance cluster file system for an ESX/ESXi cluster.
 2. **VMotion:** A service that allows for the migration of a virtual machine from one physical server to another physical server while the virtual server runs continuously and without any interruption of ongoing transactions.
 3. **Storage VMotion:** A product that can migrate files from one datastore to another datastore while the virtual machine that uses the datastore continues to run.
 4. **Virtual SMP:** A feature that allows a virtual machine to run on two or more physical processors at the same time.

VMware vSphere

5. Distributed Resource Scheduler (DRS):

- A system for provisioning virtual machines and load balancing processing resources dynamically across the different physical systems that are in use.
- A part of the DRS called the Distributed Power Management (DPM) module can manage the power consumption of systems.

5. vNetwork Distributed Switch (DVS):

- A capability to maintain a network runtime state for virtual machines as they are migrated from one physical system to another.
- DVS also monitors network connections, provides firewall services, and enables the use of third-party switches such as the Cisco Nexus 1000V to manage virtual networks.

Understanding Machine Imaging

Understanding Machine Imaging

- A system image makes a copy or a clone of the entire computer system inside a single container such as a file.
- The system imaging program is used to make this image and can be used later to restore a system image.
- Some imaging programs can take snapshots of systems, and most allow you to view the files contained in the image and do partial restores.
- A prominent example of a system image and how it can be used in cloud computing architectures is the Amazon Machine Image (AMI) used by Amazon Web Services to store copies of a virtual machine.
- Porting Applications – Cloud applications can be moved between virtual systems to meet demand, but transferring them between cloud vendors is challenging due to the lack of interoperability between major providers.

PORTING APPLICATIONS

3. VAA Implementation –

- VAAs act as self-contained application images, managed through AppZero tools.
- They enable applications to be installed on Windows Servers without interference, improving portability and reducing platform dependence.

4. Stateless Cloud and Efficiency –

- VAAs can segment applications into read-only and modifiable components, reducing data replication needs.
- The stateless cloud model stores application state separately, enabling flexible deployment across cloud systems while minimizing stored system images.

PORTING APPLICATIONS

1. The Simple Cloud API –

- If you build an application on a platform such as Microsoft Azure, porting that application to Amazon Web Services or GoogleApps may be difficult, if not impossible.
- In an effort to create an interoperability standard, Zend Technologies has started an open source initiative to create a common application program interface that will allow applications to be portable.
- The initiative is called the Simple API for Cloud Application Services. It has as its goal a set of common interfaces for:
 1. **File Storage Services:** Currently Amazon S3, Windows Azure Blob Storage, Nirvanix, and Local storage is supported by the Storage API.
 2. **Document Storage Services:** Amazon SimpleDB and Windows Azure Table Storage are currently supported.
 3. **Simple Queue Services:** Amazon SQS, Windows Azure Queue Storage, and Local queue services are supported.

PORTING APPLICATIONS

2. AppZero Virtual Application Appliance—

- Applications running in data centers are tightly coupled with the operating system and hardware. Moving an application is not as simple as transferring a machine image.
- Windows applications rely on Dynamic Link Libraries (DLLs) and registry modifications. This dependency makes porting applications between platforms difficult.
- PaaS developers package the entire software stack, including the OS and application logic. This creates vendor lock-in, making it difficult to switch cloud providers.
- **AppZero's Virtual Application Appliance (VAA) Solution:**
 - ✓ AppZero provides a Virtual Application Appliance (VAA) to improve application portability.
 - ✓ The VAA acts as a Virtualization Layer between the OS and applications.
 - ✓ It mediates file I/O, memory I/O, and DLL calls, sandboxing the application.
 - ✓ Unlike traditional installations, VAAs do not modify the OS registry or system files.

PORTING APPLICATIONS

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CAPACITY PLANNING

CAPACITY PLANNING

- Capacity planning aims to **match demand with available resources** in cloud computing.
- It involves **analyzing existing infrastructure, measuring performance, and predicting future demand.**
- Understanding workloads is **crucial for proper scaling.**
- Resources like **Cpu, Memory, Storage, and Network Capacity** must be evaluated for utilization rates and bottlenecks.
- **Scaling Strategies:**
 - ✓ **Vertical Scaling (Scaling Up):** Adding more resources (CPU, memory) to a single system.
 - ✓ **Horizontal Scaling (Scaling Out):** Adding more nodes to distribute workloads efficiently

CAPACITY PLANNING

- Capacity planners determine which resource (CPU, memory, disk, or network) **limits performance and plan upgrades accordingly.**
- Network I/O is a frequent bottleneck in cloud environments.
- **System Metrics and Performance Measurement:**
 - ✓ Metrics like **Total Workload (WT)**, **Average Workload (WAVG)**, and **Peak Workload (WMAX)** are used to **analyze system performance.**
 - ✓ Tools like **RRDTool**, **Task Manager**, and **sar (Linux)** assist in **tracking these metrics**

CAPACITY PLANNING

- Systems are tested under heavy loads to determine breaking points and required adjustments.
- Predictive analysis helps in estimating future resource requirements.
- Cloud networks face variability in performance due to factors like ISP bandwidth, network congestion, and cloud provider limitations.
- Packet sniffers like Wireshark and TCPdump are used for network monitoring.
- **Cloud Cost and Resource Allocation:**
 - ✓ Capacity planning ensures efficient use of cloud resources to avoid over-provisioning and unnecessary costs.
 - ✓ Amazon EC2 instance types provide different performance levels, requiring careful selection based on workloads

CAPACITY PLANNING

Iterative Process of Capacity Planning:

- It follows a cyclical approach:
 - ✓ Analyze current system performance.
 - ✓ Measure workloads and identify resource ceilings.
 - ✓ Conduct stress testing.Predict future demand.
 - ✓ Allocate or deallocate resources accordingly
- The Capacity planning is not just about optimizing performance but ensuring that cloud infrastructure can handle demand effectively while balancing cost and efficiency.